

Alan J. Kay

SENIOR SOFTWARE ENGINEER II · MINNEAPOLIS, MN

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SUMMARY

Senior software engineer with 11+ years in platform and product engineering, delivering cloud foundations, developer experience, and AI enablement at enterprise scale. I take projects from idea to sustained adoption: building the solution, enabling the people who use it, and measuring the results. Recent impact includes company-wide AI tool rollouts (GitHub Copilot, Claude Code), cloud platform foundations supporting 3,500+ AWS accounts, and developer-productivity programs with \$10M+ annual ROI.

PROFESSIONAL EXPERIENCE

Principal Financial Group

Des Moines, IA

Sr. Software Engineer II · IS ExE PDIO (Product Delivery Intelligence & Optimization)

Jan 2025 – Present

- **AI measurement and insight.** Advanced AI measurement literacy across the ~4,000-person product delivery organization, teaching teams to distinguish AI adoption from AI impact across pull request (PR) cycle time, review time, lead time for change, and defect rate. Led a proof of concept measuring actual AI code percentage and analyzing AI sessions, both to guide model selection and to help individuals improve how they write requirements, steer the agent, scope work, and know when to stop.
- **Claude Code rollout.** Co-led the rollout: authored all user documentation, provided troubleshooting and support, and enabled Claude Code platform owners to upskill by sharing how GitHub Copilot is managed. Reached ~800+ weekly users within months and produced insight into where Copilot versus Claude Code excels.
- **DX platform rollout.** Prototyped, vetted via a 250-person proof of concept (POC), and rolled out DX (getdx.com) to ~4,000 product-delivery contributors, giving teams a data-driven way to surface their own wins and pain points and drive change through a five-step process (Snapshot, Triage, Review, Prioritize, Measure). Drove a cultural shift toward team-owned improvement, lifting all measured drivers versus onboarding baselines with a conservative ~\$10M annual ROI.

Sr. Software Engineer II · IS ExE EngEx (Engineer Experience)

Sep 2022 – Jan 2025

- **GitHub Copilot rollout.** Led the effort from prototype to company-wide rollout: a 250-person trial, business case to senior leadership, and partnership with legal, security, and risk for Principal's first AI tool. The trial proved 6%+ productivity gain against a 0.5% break-even; Copilot now serves ~3,500 weekly users, returning ~\$26M in annual productivity. Delivered training, demos, and a tech-conference session to 3,000+ engineers.
- **Engineering data pipeline.** Designed and built a pipeline feeding the company data lake (Snowflake) with curated data products, processing ~2M records/day synchronously and asynchronously at minimal overhead. Data supplied from this pipeline powers cloud security posture and misconfiguration alerts and finance billback of AI licensing.
- **Engineer workstation setup utility.** Built an engineer-first utility (Ansible; Windows and Linux via WSL) that reduced new-engineer setup from multiple business days to under an hour. Run by ~5,700 unique users in a year, saving ~91,000 engineer hours (~\$6.8M in recovered productivity).
- **License management modernization.** Cut excess MSDN and IBM RAD licensing for \$150K+ in annual savings and migrated the on-premises JetBrains license server to a cloud offering for scale and availability.

Sr. Software Engineer · IS CPE (Cloud Platform Engineering)

Apr 2021 – Sep 2022

- **Cloud landing zones and account vending.** Built and evolved Principal's AWS, GCP, and Azure cloud foundations, now supporting 3,500+ AWS accounts with new accounts vended in under an hour and foundational changes rolled out at scale. AWS became the preferred platform over on-premises alternatives.
- **IaC pre-deploy scanning.** Created a scanning solution (Checkov and custom checks), now required in Principal's production pipeline: ~2,700 runs/day across ~4,000 repositories, evaluating ~300 rules per run.

Prevented production regressions and materially improved cloud security posture.

- **Software delivery platforms via IaC.** Administered critical delivery platforms (GitHub, Azure DevOps) for 3,500+ engineers through infrastructure as code and automation, enabling cloud delivery while meeting regulatory and security controls.

Software Engineer · PGI Talent & Organizational Competency

Mar 2019 – Apr 2021

- **Technical interviewing via pair programming.** Assessed engineering candidates, improving new-hire quality and retention.
- **Lean engineering practice and consulting.** Coached teams through pairing, mobbing, and consulting to raise product quality.

Application Developer · PGI DESTination Program

May 2014 – Mar 2019

- **DESTination program future vision.** Key contributor to a program that redefined the business unit's technological future vision and the steps to reach it. Designed and delivered high-performing, high-impact systems in support of that vision.
- **Loosely coupled integration framework.** Designed and implemented a framework that lets systems be added, replaced, or removed with agility.

Earlier roles: Data Management Intern and Application Development Intern, Principal Financial Group (2012–2013); Systems Administrator Assistant and Support Technician, Drake University (2011–2014); Design and Engineering Intern and Warehouse Employee, Maclean Power Systems (2010–2011).

SKILLS

AI & PRODUCTIVITY AI-assisted development across the SDLC · AI tooling: evaluation, business case, administration & governance · AI enablement: training, documentation, pairing · AI impact measurement & analysis · GitHub Copilot · Claude Code · SPACE metrics · DORA metrics · DX (getdx.com) · Developer experience & productivity

PLATFORMS Amazon Web Services (S3, SQS & SNS, Lambda, CloudWatch) · Google Cloud Platform (GCP) · Microsoft Azure · GitHub · Azure DevOps · Snowflake · PostgreSQL · Microsoft SQL Server · Mule ESB

LANGUAGES Python · TypeScript · C# (MVC and Web API) · PowerShell · Java (limited) · Visual Basic.NET (limited)

INFRASTRUCTURE & TOOLING Terraform · CloudFormation · AWS CDK · Ansible · Checkov · Windows Subsystem for Linux (WSL) · Workstation setup automation · Docker

PRACTICES Platform engineering · Cloud security posture · Cost optimization & license management · Data pipelines · DevOps & CI/CD · Lean engineering · Technical leadership · Enablement & documentation

EDUCATION

Drake University

Des Moines, IA

Bachelor of Science in Business Administration

2014

- Information Systems and Marketing Majors, Computer Science Minor
- Cumulative GPA 3.67/4.0 – Cum Laude, Departmental Honors

LEADERSHIP & RECOGNITION

- **Rebuilt trust with a chief accounting officer.** Led a team to deliver a product I defined, on time and defect-free, while training the engineers on AWS and industry-leading practices, rebuilding a chief accounting officer's trust in IT.
- **Praised for enterprise-wide improvement.** Repeatedly recognized by the director of engineering for improving the enterprise while addressing individual teams' needs.